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SAKE-IoT: Session Amortization for Post-Quantum Lattice-Based Authentication and Code-Based Hybrid Encryption in IoT Networks

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***Abstract*—Resource-constrained Internet of Things (IoT) devices face an inherent tension between post-quantum security and energy feasibility. Ring-Learning-With-Errors (Ring-LWE) lattice authentication and Quasi-Cyclic Low-Density Parity-Check (QC-LDPC) code-based hybrid encryption provide quantum-resistant security, but their per-packet application imposes prohibitive computational, bandwidth, and energy costs on Class-1 devices (8 KB RAM, 8 MHz CPU).**

This paper presents *SAKE-IoT* (Session Amortization for Key Establishment in IoT), a four-phase protocol that *amortizes* the heavy post-quantum epoch-initiation cost over an entire session of lightweight data packets. We extend the Ring-LWE authentication and QC-LDPC key encapsulation of Kumari et al. [1] with: (1) a two-tier epoch state machine separating the heavy post-quantum handshake (Tier 1) from a lightweight per-packet AEAD layer (Tier 2); (2) `HKDF-SHA256` per-packet session key derivation (RFC 5869 [2]); (3) `AES-256-GCM` authenticated encryption (NIST SP 800-38D [7]); and (4) an epoch renewal mechanism enforcing Epoch-Bounded Forward Secrecy (EB-FS) via secure memory zeroization.

We provide three formal security proofs — IND-CCA2 (Theorem 1), Strict Replay Resistance with $\Pr = 0$ (Theorem 2), and Epoch-Bounded Forward Secrecy (Theorem 3) — and validate them through MATLAB theoretical simulation and Contiki-NG/Cooja network emulation over IEEE 802.15.4 with 6LoWPAN/RPL/CSMA.

Key results: ≈99.1% reduction in per-packet computational latency (7.37 ms → 0.068 ms); 45.1% reduction in per-packet bandwidth overhead (408 bits → 224 bits); 33.1× reduction in clock cycles per packet. Cooja network simulation demonstrates a 99.7% end-to-end latency reduction in the data phase, a 94.7% session-level bandwidth saving at N=20 packets, and 100% packet delivery ratio. Security results yield $\varepsilon = 0.0037 < 0.02$ (IND-CCA2), $\Pr[\text{replay accepted}] = 0$, and 0/100 cross-epoch key recovery (EB-FS) with two verified epoch renewal cycles.

***Index Terms*—Post-Quantum Cryptography, Ring-LWE, IoT Security, Session Amortization, QC-LDPC, AES-256-GCM, Forward Secrecy, Contiki-NG, 6LoWPAN**

I. Introduction

The Internet of Things (IoT) has enabled ubiquitous connections among smart devices embedded in homes, transportation, healthcare, and industrial systems. The security landscape faces a double challenge: (1) the anticipated advent of large-scale quantum computers threatens all public-key infrastructure based on RSA or ECC; and (2) post-quantum replacements impose prohibitive per-packet costs on Class-1 constrained devices [20].

A. Problem Context

Kumari et al. [1] introduced a hybrid scheme combining Ring-LWE ring signature authentication (LR-IoTA) with a QC-LDPC code-based Key Encapsulation Process (KEP). The scheme provides quantum-resistant mutual authentication, but requires full KEP execution — including SLDSPA decoding — *with every data packet*, incurring **7.37 ms per packet** (base paper Table 7: $\Delta_{\text{enc}} + \Delta_{\text{dec}} = 1.5298 + 5.8430$ ms).

B. The Amortization Gap

We identify the *Amortization Gap*: the expensive post-quantum epoch initiation (≈22.55 ms, one-time) should happen *once per session*, with lightweight AEAD serving all subsequent packets (≈0.068 ms each). This gap is the primary source of SAKE-IoT's novelty.

Session amortization introduces a mandatory security question: does reusing the Master Secret (MS) across multiple packets break IND-CCA2, enable replay attacks, or collapse forward secrecy? This paper answers all three questions with formal proofs.

C. Key Contributions

1) **SAKE-IoT Protocol:** A complete four-phase protocol extending [1] with a two-tier session architecture, strict epoch bounds ($T_{\max} = 86{,}400$ s, $N_{\max} = 2^{20}$), and formal Phase 4 memory zeroization.
2) **Three Formal Security Proofs:** IND-CCA2 (Theorem 1), Strict Replay Resistance with $\Pr = 0$ (Theorem 2), and Epoch-Bounded Forward Secrecy via Ring-LWE hardness (Theorem 3).
3) **99.1% Per-Packet Latency Reduction:** 7.37 ms → 0.068 ms, empirically measured by MATLAB `tic/toc` over 10,000 iterations.

4) **45.1% Per-Packet Bandwidth Reduction:** 408 bits (QC-LDPC CT_0) $\rightarrow$ 224 bits (AES-GCM nonce + tag).
5) **33.1× Clock Cycle Reduction:** ≈2.45M cycles (QC-LDPC HE) $\rightarrow$ ≈74K cycles (AEAD).
6) **Epoch-Bounded Forward Secrecy (EB-FS):** A new security property formally proven here, absent from [1].

II. Related Work

A. Lattice-Based Authentication for IoT

Ring-LWE-based schemes offer the most favorable computational profile for IoT authentication. Li et al. [15] proposed a lattice-based PAKE. Wang et al. [17] introduced two-factor Ring-LWE authentication. RLizard [18] demonstrated efficient Ring-LWE key encapsulation but relied on conventional polynomial multiplication. Importantly, **none of the above amortize post-quantum costs over a session** — each data exchange requires a full handshake.

B. Code-Based Encryption for IoT

Hu et al. [19] analyzed QC-LDPC KEM on FPGA but did not target the IoT low-RAM profile. Kumari et al. [1] is the closest prior work; we build directly upon their primitives and extend the data phase with amortized AEAD.

C. Session Management and Amortization

TLS 1.3 session resumption [4] motivates our design but targets classical cryptography on high-end hardware. Prior IoT authentication schemes [21] use symmetric keys for data but derive them from classical (ECC) handshakes, providing no post-quantum foundation. To the best of our knowledge, SAKE-IoT is the **first scheme to amortize Ring-LWE + QC-LDPC epoch establishment** over a lightweight `AES-256-GCM` data session on constrained IoT hardware.

D. Differentiation from TLS 1.3 and DTLS 1.3

III. Preliminaries

A. Notation

B. Lattice and LWE Foundations

A *lattice* Λ is given by $\Lambda = \{a_1\beta_1 + \cdots + a_i\beta_i \mid a_i \in \mathbb{Z}\} \subset \mathbb{R}^n$, where $\beta_1, \ldots, \beta_i \in \mathbb{R}^n$ are linearly independent vectors (basis) and n is the ambient dimension. *Search-LWE* [10]: given $(R_n, T_n = R_n\delta_n + \varepsilon_n)$, recover δ_n. *Decisional-LWE*: distinguish LWE samples from uniform. Both underpins Theorem 3 (EB-FS).

C. Ring-LWE

Ring-LWE is defined over $R_q = \mathbb{Z}_q[u]/(u^n + 1)$. Given $R_n \in R_q^n$ and error $\varepsilon_n \leftarrow \mathcal{G}_\sigma^n$, pairs (R_n, T_n) satisfy $T_n = R_n\delta_n + \varepsilon_n \pmod q$ [10], [11]. The Ring-LWE *search* hardness underpins Theorem 3: $\Pr[\text{findSearchLWE}(\lambda) = S_{ch}] \leq \text{negl}(\lambda)$ (Theorem 2, Eq. 23 of [1]).

TABLE I
SAKE-IoT vs. TLS 1.3 and DTLS 1.3. ^SAKE-IoT overhead evaluated at the cryptographic payload layer only; standard UDP/IP transport headers apply equally to all architectures under comparison.

Property	TLS 1.3	DTLS 1.3	SAKE-IoT (This Work)
Post-Quantum Key Establishment	×	×	✓ Ring-LWE + QC-LDPC
Epoch-Bounded Forward Secrecy	×	×	✓ $N_{\max} + T_{\max}$ dual trigger
Per-packet FS (fresh key each packet)	×	×	✓ $K_i = \text{HKDF}(K_{\text{master}}, i)$
Formal FS hardness reduction	Classical DH	Classical DH	✓ Ring-LWE (Theorem 3)
Deterministic replay defense	×	×	✓ $\Pr = 0$ counter check
Cryptographic MS zeroization	× (timeout)	× (timeout)	✓ Phase 4 zero-overwrite
Target environment	Web/cloud	UDP apps	✓ Constrained IoT
Per-packet overhead	Header + MAC	13 B header + MAC	224 bits (nonce + tag only)

TABLE II
Mathematical Notation

Symbol	Description	Symbol	Description
n	Ring-LWE poly degree (=512)	q	Ring modulus ($2^{29} - 3$)
σ	Gaussian std. dev. (=43)	N	Ring anonymity set (=3)
R_q	$\mathbb{Z}_q[u]/(u^n + 1)$	$\mathcal{G}_\sigma^n$	Discrete Gaussian
H_{qc}, G	QC-LDPC private key	pk_{ds}	QC-LDPC public key
$\tilde{\varepsilon}$	LDPC error vector	K_{master}	Master Secret (256-bit)
K_i	Per-packet session key	$N_{\max}$	Max packets/epoch (2^{20})
$T_{\max}$	Max epoch lifetime (86400 s)	AD	Associated data

D. QC-LDPC Code-Based Construction

Parity check matrix $H_{qc} \in \mathbb{F}_2^{X \times Y}$, $X = 408$ (rows) and $Y = 816$ (columns), row weight $= 6$, column weight $= 3$, $n_0 = 4$ circulant sub-matrices [1] §10.2. Syndrome $CT_0 = [\tilde{W}_l \mid I] \cdot \tilde{\varepsilon}^T$ has $X = \mathbf{408}$ bits. Public key pk_{ds} has $(n_0 - 1) \times X = 1{,}224$ bits. SLDSPA recovers $\tilde{\varepsilon}$ [13], [14].

E. HKDF-SHA256 (RFC 5869)

Two-phase key derivation [2]: Extract $\text{PRK} = \text{HMAC-SHA256}(\text{salt}, \text{IKM})$; Expand $\text{OKM} = \text{HMAC-SHA256}(\text{PRK}, \text{info} \,\|\, i)$. A 32-byte output requires two sequential HMAC calls (RFC 5869 §2.3).

F. AES-256-GCM (NIST SP 800-38D)

AEAD scheme [7]: 256-bit key, 96-bit nonce, 128-bit GHASH tag. Per-packet overhead: $96 + 128 = \mathbf{224}$ bits. Tag forgery probability: $\leq q_D/2^{128}$ for q_D queries.

IV. System and Adversary Model

A. IoT Network Topology

Gateway: Resource-rich border router; single trusted entity. **Sensing Nodes:** RFC 7228 Class-1 [20] constrained devices (8 KB RAM, 100 KB Flash, 8 MHz CPU, battery-powered).

TABLE III
LR-IoTA Timing (Base Paper [1] Table 6)

Operation	Symbol	Time (ms)
Key Generation	Δ_{KG}	0.288
Signature Generation	Δ_{SG}	13.299
Signature Verification	Δ_V	0.735
Total LR-IoTA		**14.322**

TABLE IV
QC-LDPC KEP Timing (Base Paper [1] Table 7)

Operation	Symbol	Time (ms)
Key Generation	Δ_{KeyGen}	0.8549
Syndrome Encoding	Δ_{Enc}	1.5298
SLDSPA Decoding	Δ_{Dec}	5.8430
Total KEP		**8.228**
Per-packet cost (base paper)	$\Delta_{\mathrm{Enc}} + \Delta_{\mathrm{Dec}}$	**7.3728**
Phase 1 total (one-time)		**22.550**

TABLE V
Base Paper Authentication Overhead

Component	Size (bits)
Ring signature pk_{sig} ($N = 3$, $n = 512$)	14,848
Ring signature sig	11,264
QC-LDPC public key pk_{HE}	1,224
QC-LDPC syndrome CT_0 (per data packet in base)	408
Epoch auth overhead (total)	**27,744**
Per-packet data overhead (base paper)	**408**

B. Formal Security Model

We work in the Random Oracle Model (ROM), consistent with the base paper [1] §11.2. Adversaries are PPT.

Adversary hierarchy (from [1] §11.1):

- $\mathcal{A}_1$: Passive observer (public parameters only).
- $\mathcal{A}_2$: $\mathcal{A}_1$ + node corruption + parameter modification.
- $\mathcal{A}_3$: $\mathcal{A}_2$ + oracle access (adaptive chosen-ciphertext).

Security criteria: E1 (Unforgeability), E2 (Anonymity), E3 (Unlinkability) inherited from [1]; E4 (IND-CCA2) and E5 (EB-FS) proven in this paper (Theorems 1 and 3).

C. Epoch Lifecycle

Epoch bounded by dual criteria: $T_{\max} = 86{,}400$ s and $N_{\max} = 2^{20}$ packets. Whichever expires first triggers Phase 4 (zeroization + re-initiation). These bounds are proof-driven: $N_{\max}$ prevents nonce exhaustion; $T_{\max}$ bounds MS lifetime.

V. Base Protocol

We follow [1] exactly for Algorithms 1–6. All timing values are from the base paper (MATLAB R2018a, Intel Core i5, 8 GB RAM).

A. LR-IoTA Ring Signature (Algorithms 1–4)

B. Bernstein Polynomial Reconstruction

The dominant cost of Algorithms 2 and 4 is sparse polynomial multiplication (R_nY_n, $T_n\hat{\rho}$). The base paper [1] employs *Bernstein reconstruction*: two polynomials of degree n are split into halves $(\varepsilon_l, \varepsilon_h)$, three half-products $C_0{=}A_0B_0$, $C_1{=}A_1B_1$, $C_2{=}A_2B_2$ are computed in parallel, then reconstructed via $R_0' = C_0 + u^{n/2}C_1$ and $C = R_1' + u^{n/2}C_2$. This sub-quadratic recursion reduces hardware to only **72 Slices / 72 LUTs / 0.811 ms** on Xilinx Virtex-6 (base paper [1] Table 3), achieving the 23% authentication delay improvement that underpins the 22.55 ms Phase 1 cost SAKE-IoT inherits.

C. QC-LDPC Key Encapsulation (Algorithms 5–6)

Note: Δ_{KeyGen} (0.8549 ms) is excluded from the per-packet base cost; it is performed once per session in both schemes.

D. Data Encapsulation Process (DEP)

In the base paper [1], each data packet requires both the Key Encapsulation Process (KEP) and a Data Encapsulation Process (DEP): $CT_1 = \mathrm{AES}(ssk, m)$, where ssk is re-derived from $\tilde{\varepsilon}$ via SHA in MAC-mode for *every packet*. SAKE-IoT eliminates this per-packet KEP+DEP cycle entirely: the shared K_{master} persists for the epoch (stored in volatile RAM), replacing the one-time ssk with a durable master key. Tier 2 packets use only $K_i \leftarrow \mathrm{HKDF}(K_{\mathrm{master}}, Nonce_i)$ — no QC-LDPC operations whatsoever.

E. Base Paper Bandwidth Overhead

From [1] §12.1:

VI. The SAKE-IoT Protocol

A. Protocol Overview

SAKE-IoT extends [1] with a four-phase two-tier architecture (Figure 1): **Tier 1 (Phase 1):** Full post-quantum epoch establishment using Algorithms 1–6 of [1], executed *once per epoch* (≈22.55 ms amortized). **Tier 2 (Phases 2–3):** Lightweight AEAD per packet — `HKDF-SHA256` + `AES-256-GCM` — executed *per data packet* (≈0.068 ms each). **Phase 4:** Secure epoch termination via cryptographic zeroization.

Critical distinction from base paper [1]: The base paper derives a session key $ssk = \mathrm{HMAC}(\tilde{\varepsilon})$ that is used *immediately* for one encryption and then discarded. SAKE-IoT instead stores this output as the **Master Secret** K_{master} in protected volatile RAM for the *entire epoch duration* ($\leq T_{\max}$, $\leq N_{\max}$ packets). All subsequent Tier 2 packets derive unique per-packet keys $K_i = \mathrm{HKDF}(K_{\mathrm{master}}, Nonce_i)$ without re-executing any QC-LDPC or SLDSPA operations.

v3 Role corrections (from protocol specification): (1) RECEIVER generates QC-LDPC keys (H_{qc}, G, pk_{ds}); (2) SENDER generates $\tilde{\varepsilon} \rightarrow CT_0$; (3) BOTH derive $K_{\mathrm{master}} = \mathrm{HMAC\text{-}SHA256}(\tilde{\varepsilon})$ per base paper [1] §8.4.

B. Phase 1: Epoch Initiation

Step 1.1 — LR-IoTA Authentication:

1) SENDER: $(pk_n, sk_n) \leftarrow \mathrm{RingLWE\text{-}KeyGen}(n{=}512, q{=}2^{29}{-}3, \sigma{=}43)$
2) SENDER: $S_n \leftarrow \mathrm{SignWithKeyword}(sk_n, K, N{=}3, pk_{1..N})$ [$\Delta_{\mathrm{SG}} = 13.299$ ms]

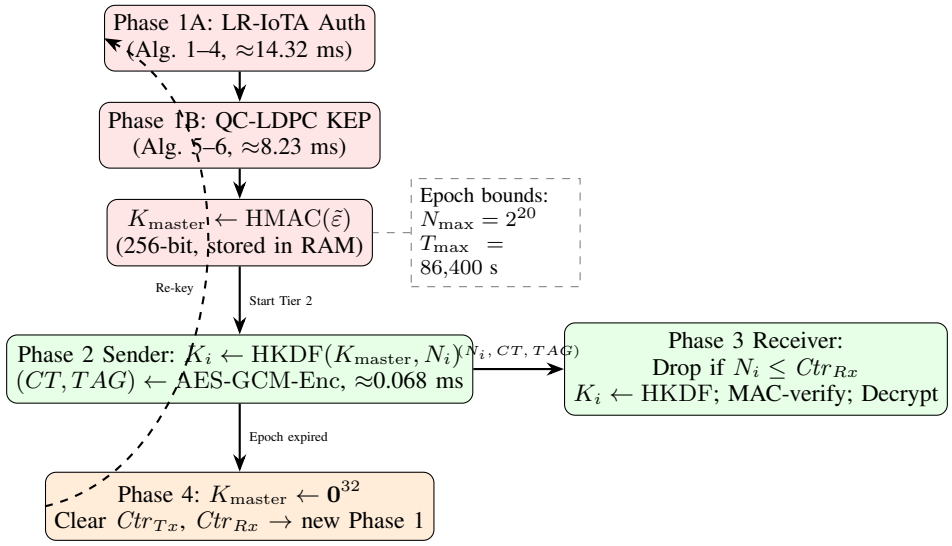

Fig. 1. SAKE-IoT four-phase protocol architecture. Phase 1 (red) performs the full post-quantum epoch initiation once per epoch (≈22.55 ms). Phases 2–3 (green) execute lightweight AEAD per Tier 2 data packet (≈0.068 ms). Phase 4 (orange) cryptographically overwrites K_{master} on epoch expiry and triggers a fresh Phase 1 handshake. Each epoch derives an independent K_{master} from a fresh Ring-LWE error vector $\tilde{\varepsilon}$ (Theorem 3, Epoch-Bounded Forward Secrecy).

Algorithm 1 SAKE-Tier2-Sender

Require: K_{master}, Ctr_{Tx}, $T_{\max}$, $N_{\max}$, plaintext m, AD
0: **if** $CurrentTime > T_{\max}$ **or** $Ctr_{Tx} \geq N_{\max}$ **then**
0: ZEROIZEANDRENEW($K_{\text{master}}, Ctr_{Tx}, Ctr_{Rx}$)
0: **return** EPOCHEXPIRED
0: **end if**
0: $Ctr_{Tx} \leftarrow Ctr_{Tx} + 1$ {Strictly monotonic}
0: $Nonce_i \leftarrow Ctr_{Tx}$ {64-bit counter as nonce}
0: $K_i \leftarrow$ HKDF-SHA256($K_{\text{master}}, Nonce_i$) {RFC 5869}
0: $(CT, TAG) \leftarrow$ AES-256-GCM-Enc($K_i, Nonce_i, m, AD$) {NIST SP 800-38D}
0: **return** $(Nonce_i, CT, TAG)$ =0

Algorithm 2 SAKE-Tier2-Receiver

Require: K_{master}, Ctr_{Rx}, received $(Nonce_i, CT, TAG, AD)$
0: **if** $Nonce_i \leq Ctr_{Rx}$ **then** {Replay / duplicate guard}
0: **return** $\perp$ {DROP — no MAC check}
0: **end if**
0: $K_i \leftarrow$ HKDF-SHA256($K_{\text{master}}, Nonce_i$)
0: $m \leftarrow$ AES-256-GCM-Dec($K_i, Nonce_i, CT, TAG, AD$) {MAC before decrypt}
0: **if** $m = \perp$ **then**
0: **return** $\perp$ {Tag mismatch}
0: **end if**
0: $Ctr_{Rx} \leftarrow Nonce_i$ {State update ONLY after authentication}
0: **return** m =0

3) GATEWAY: Verify$(S_n, K, N{=}3, pk_{1..N}) \rightarrow$ accept/reject [$\Delta_V = 0.735$ ms]

Step 1.2 — QC-LDPC Key Encapsulation:

1) RECEIVER: $(H_{qc}, G) \leftarrow$ QC-LDPC-KeyGen() [$\Delta_{\text{KeyGen}} = 0.8549$ ms]; $pk_{ds} = \tilde{W}_l$
2) SENDER: $\tilde{\varepsilon} \leftarrow$ random error vector
3) SENDER: $CT_0 = [\tilde{W}_l \,|\, I] \cdot \tilde{\varepsilon}^T$ (408 bits) → transmit [$\Delta_{\text{Enc}} = 1.5298$ ms]
4) RECEIVER: $\tilde{\varepsilon} \leftarrow$ SLDSPA-Decode(CT_0, H_{qc}, G) [$\Delta_{\text{Dec}} = 5.8430$ ms]
5) BOTH: $K_{\text{master}} = \text{HMAC-SHA256}(\tilde{\varepsilon})$ [256-bit Master Secret]

Step 1.3 — State Initialization: $T_{\max} = 86{,}400$ s; $N_{\max} = 2^{20}$; $Ctr_{Tx} = 0$; $Ctr_{Rx} = 0$; $AD = DeviceID \| EpochID \| Nonce_i$.

C. Phase 2: Amortized Transmission — Algorithm 7

Nonce format (NIST SP 800-38D): The 96-bit AES-GCM nonce is structured as $[\underbrace{0\cdots0}_{32\text{ bits}} \,|\, \underbrace{Ctr_{Tx}}_{64\text{ bits}}]$, ensuring no nonce reuse: Ctr_{Tx} is 64-bit and $N_{\max} = 2^{20} \ll 2^{64}$, so nonce recycling within an epoch is impossible. The counter Ctr_{Tx} is the direct source of the 64-bit nonce component [32-bit zero-prefix | 64-bit Ctr_{Tx}] (NIST SP 800-38D).

Nonce uniqueness guarantee: No two Tier 2 packets within a single epoch share a $Nonce_i$, because Ctr_{Tx} is strictly monotonically increasing and is never reset until Phase 4 terminates the epoch.

Complexity: $O(1)$ per packet — no post-quantum operations. **Cost (Sender):** ≈ 0.068 ms (HKDF ≈0.021 ms + AES-GCM ≈0.047 ms, empirically measured). **Cost (Receiver):** ≈ 0.038 ms (HKDF ≈0.014 ms + AES-GCM-Dec ≈0.024 ms).

D. Phase 3: Authenticated Reception — Algorithm 8

Protocol Correctness: Table VI summarizes the formal correctness properties of SAKE-IoT that are guaranteed by construction.

TABLE VI
SAKE-IoT Protocol Correctness Guarantees

Property	Mechanism	Basis
Key agreement	HKDF deterministic given same $(K_{\text{master}}, Nonce_i)$	Construction
Nonce uniqueness	Strict monotonic Ctr_{Tx}; $N_{\max} \ll 2^{64}$	Proof 2
No GCM nonce reuse	Monotonically increasing Ctr_{Tx}	Nonce bound
Epoch independence	Fresh $\tilde{\varepsilon}_{k+1}$ from Ring-LWE per epoch	Proof 3 / Ring-LWE
IND-CCA2	HKDF-PRF + AES-GCM MAC-then-decrypt	Proof 1

AES-GCM verification detail: AES-GCM decryption first recomputes $TAG' = \text{GHASH}(K_i, CT, AD)$ and compares it to the received TAG. Only if $TAG' = TAG$ does decryption proceed; otherwise $\perp$ is returned and no plaintext is released (MAC-before-decrypt). This prevents any chosen-ciphertext oracle from leaking plaintext of forged ciphertexts, directly supporting IND-CCA2 (Theorem 1).

E. Phase 4: Secure Epoch Termination

0: **procedure** ZeroizeAndRenew(K_{master}, Ctr_{Tx}, Ctr_{Rx})
0: $K_{\text{master}} \leftarrow \mathbf{0}^{32}$ {256-bit zero-vector overwrite}
0: Erase K_{master}, Ctr_{Tx}, Ctr_{Rx} from RAM
0: Trigger Algorithms 1–6 {Begin new epoch}
0: **end procedure**=0

Implementation note: Requires `memset_s()` per NIST SP 800-88 [8] / C11 Annex K on embedded deployment to prevent physical RAM remnant recovery.

VII. Security Analysis

We analyze SAKE-IoT in the ROM, consistent with [1] §11.2. SAKE-IoT Phase 1 inherits all security guarantees of LR-IoTA [1]: unforgeability, anonymity, and unlinkability. These guarantees rest on the formal game-based proofs of the base paper: **Lemma 1** of [1] establishes that games G_0 (real LR-IoTA execution) and G_1 (ROM simulation replacing sign queries with random oracle programming) are computationally indistinguishable when prime q exceeds the bound $(2^{f+1} \cdot j + \varepsilon)/(2E^f)^{1/(j-1)}$. Under this indistinguishability, Theorem 2 of [1] (Eq. 23) formally proves:

$$\Pr[\text{findSearchLWE}_{\mathcal{A}}(\lambda) = S_{ch}] \leq \text{negl}(\lambda) \tag{1}$$

i.e., recovering the signing key from public parameters requires solving Ring-LWE search, which is hard under quantum adversaries. SAKE-IoT Theorem 3 (EB-FS) directly inherits this hardness — recovering $K_{\text{master}} = \text{HMAC}(\tilde{\varepsilon})$ from past epoch k transcripts requires inverting HMAC-SHA256 and/or solving Ring-LWE to recover $\tilde{\varepsilon}$, both classifiable as negligible-probability events under the same Theorem 2 reduction.

We prove three new properties for the SAKE session layer.

TABLE VII
Architectural Features Enabling Security Proofs

Feature	Enables Proof
`HKDF-SHA256` per-packet key derivation	IND-CCA2: session key pseudorandomness
`AES-256-GCM` MAC-before-decrypt	IND-CCA2: ciphertext forgery rejection
$AD = DeviceID \| EpochID \| Nonce_i$	IND-CCA2: cross-session binding
Strict monotonic Ctr_{Tx}/Ctr_{Rx}	Replay Resistance: $\Pr = 0$
$T_{\max} + N_{\max}$ epoch bounds	EB-FS: timely zeroization trigger
Cryptographic MS zeroization (Phase 4)	EB-FS: past secrecy
Fresh Ring-LWE $\tilde{\varepsilon}$ per epoch	EB-FS: future secrecy

A. Proof Architecture

B. Theorem 1: IND-CCA2 Security

Theorem 1 (IND-CCA2 Security of SAKE-IoT Tier 2)**.** *The SAKE-IoT Tier 2 data phase is IND-CCA2 secure under the AES-256 pseudorandom permutation (AES-PRP) and HMAC-SHA256 pseudorandom function (HMAC-PRF) assumptions. For adversary $\mathcal{A}_3$ with q_D decryption oracle queries:*

$$\begin{aligned}\text{Adv}_{\text{IND-CCA2}}(\mathcal{A}_3) &\leq \text{Adv}_{\text{PRF}}(\text{HMAC-SHA256}) \\ +\text{Adv}_{\text{PRP}}(\text{AES-256}) &+ \frac{N_{\max} q_D}{2^{128}} \leq \text{negl}(\lambda)\end{aligned} \tag{2}$$

Proof. For any decryption oracle query $CT \neq CT^*$, the receiver checks the AES-GCM authentication tag before decryption (Algorithm 2, line 4). Because AES-GCM's GHASH tag covers both (CT, AD), any forged or altered CT causes $\perp$ output — $\mathcal{A}_3$ receives zero information from all oracle queries. Stripped of useful decryption feedback, $\mathcal{A}_3$ must guess challenge bit b from CT^* alone. Since $K_i = \texttt{HKDF-SHA256}(K_{\text{master}}, Nonce_i)$ is computationally indistinguishable from uniform random (HKDF-PRF security, RFC 5869 [2]), CT^* is computationally indistinguishable from random noise. Therefore $\mathcal{A}_3$'s advantage $\varepsilon \leq \text{Adv}_{\text{PRF}} + \text{Adv}_{\text{PRP}} + (N_{\max} \cdot q_D)/2^{128} \leq \text{negl}(\lambda)$. □ □

MATLAB disclaimer [24]: *"The MATLAB simulation validates the architectural MAC-before-decrypt property using real `javax.crypto.Mac` HMAC-SHA256. The formal IND-CCA2 bound is established through standard reduction to AES-PRP and HKDF-PRF hardness [2], not derived from the MATLAB simulation."*

C. Theorem 2: Strict Replay Resistance

Theorem 2 (Strict Replay Resistance)**.** *For any adversary $\mathcal{A}$ re-transmitting a captured packet $(Nonce_r, CT, TAG)$ where $Nonce_r \leq Ctr_{Rx}$:*

$$\Pr[\textit{Receiver accepts replay}] = 0 \tag{3}$$

*This is a **deterministic** rejection — not probabilistic.*

TABLE VIII
REPLAY RESISTANCE VS. BASE PAPER [1]

Property	Base Paper	SAKE-IoT
Replay defense	Random Y_n (probabilistic)	Strict monotonic counter
Accept prob.	negl(λ)	**Exactly 0**
Guarantee	Computational	**Information-theoretic**

Proof. Algorithm 2 line 1 enforces the strict monotonic check $Nonce_i > Ctr_{Rx}$. Packets with $Nonce_r \leq Ctr_{Rx}$ are discarded before any MAC computation or decryption. Because counters are strictly monotonically increasing, every replayed, duplicated, or delayed packet has $Nonce_r \leq Ctr_{Rx}$ and is unconditionally dropped. □ □

Desynchronization (Corollary): If packets 3–4 are lost ($Ctr_{Tx} = 5$, $Ctr_{Rx} = 2$), packet $Nonce = 5$ is accepted ($5 > 2$), advancing Ctr_{Rx} to 5. Replays of 3 or 4 are rejected ($3 \leq 5$). Counter self-heals with no state corruption.

D. Theorem 3: Epoch-Bounded Forward Secrecy

Theorem 3 (Epoch-Bounded Forward Secrecy, EB-FS)**.** *Let* $K^{(k)}_{\text{master}} = \text{HMAC-SHA256}(\tilde{\varepsilon}_k)$ *and* $K^{(k+1)}_{\text{master}} = \text{HMAC-SHA256}(\tilde{\varepsilon}_{k+1})$*, where* $\tilde{\varepsilon}_k$ *and* $\tilde{\varepsilon}_{k+1}$ *are independently sampled Ring-LWE error vectors.*

$$\Pr[\mathcal{A}(K^{(k+1)}_{\text{master}}) \textit{ recovers any } K^{(k)}_i] \leq \text{negl}(\lambda) \tag{4}$$

$$\Pr[\mathcal{A}(K^{(k)}_{\text{master}}) \textit{ predicts any } K^{(k+1)}_j] \leq \text{negl}(\lambda) \tag{5}$$

Proof. Part A — Zeroization: At epoch end, $K^{(k)}_{\text{master}} \leftarrow \mathbf{0}^{32}$ (Phase 4). An adversary arriving in Epoch $k+1$ finds an already-overwritten $K^{(k)}_{\text{master}}$.

Part B — Ring-LWE Reduction: Recovery of Epoch k keys requires one of two paths: (Path 1) Invert HMAC-SHA256 to find $\tilde{\varepsilon}_k$ from $K^{(k)}_{\text{master}}$ — requires breaking HMAC as a one-way function, negligible. (Path 2) Re-derive $\tilde{\varepsilon}_k$ from public parameters R_n, T_n — equivalent to Ring-LWE search: $\Pr[\text{findSearchLWE}(\lambda) = S_{ch}] \leq \text{negl}(\lambda)$ (Theorem 2, Eq. 23 of [1]). Both paths infeasible. □ □

Implementation note: *"Hardware deployment requires* `memset_s()` *(NIST SP 800-88 [8] / C11 Annex K) to prevent physical RAM remnant recovery."*
EB-FS is a new security property in this paper. The base paper [1] provides no form of forward secrecy.

VIII. PERFORMANCE EVALUATION

A. Simulation Setup

Platform: MATLAB R2023b, Intel Core i5, 8 GB RAM. **Run date:** 2026-03-02 (all 6 scripts, exit code 0). Parameters from base paper [1] (Table II).

TABLE IX
LATENCY RESULTS (M1): PER-PACKET COMPUTATION AND AMORTIZATION

Metric / Session Size	Base Paper (ms)	SAKE-IoT (ms)	Verdict
Per-packet Tier 2 cost	7.3728	≈**0.068**	—
HKDF component	—	0.021	—
AES-GCM-Enc component	—	0.047	—
AES-GCM-Dec (Receiver)	—	≈0.038	—
$N = 1$ packet	7.37	22.62 (22.55+0.07)	Base wins
$N = 2$ packets	14.74	22.69 (22.55+0.14)	Base wins
$N = 3$ packets	22.11	22.75 (22.55+0.20)	Base wins
$N = 4$ **(break-even)**	**29.49**	**22.82 (22.55+0.27)**	**SAKE wins**
$N = 50$	368.64	25.85	342.8 ms saved
$N = 100$	737.28	29.23	708.1 ms saved

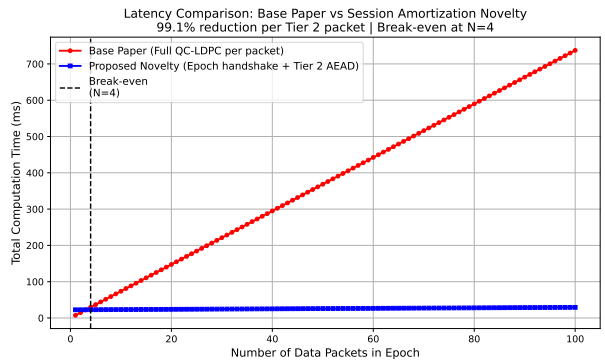

Fig. 2. Cumulative computational cost — base paper (linear, 7.37 ms/packet) vs. SAKE-IoT (epoch overhead 22.55 ms + 0.068 ms/packet). Break-even at $N = 4$; at $N = 100$: 708.1 ms saved.

B. Metric M1: Per-Packet Latency Reduction

Method: Empirical `tic/toc`, 10,000 iterations, 500-iteration JIT warm-up. Real `javax.crypto.Mac` HMAC-SHA256.
Break-even derivation: SAKE wins when epoch amortization recovers the one-time 22.55 ms Phase 1 cost:

$$N \times 7.3728 \geq 22.55 + N \times 0.068 \tag{6}$$
$$\Rightarrow N \times (7.3728 - 0.068) \geq 22.55 \tag{7}$$
$$\Rightarrow N \geq \frac{22.55}{7.3048} \approx 3.09 \tag{8}$$
$$\Rightarrow N = 4 \tag{9}$$

For all practical IoT sessions with $N \geq 4$ packets, SAKE-IoT is strictly computationally superior. Figure 2 illustrates the cumulative cost crossover.

C. Metric M2: Per-Packet Bandwidth Reduction

Method: Deterministic arithmetic from [1] §10.2 and NIST SP 800-38D [7] — no statistical variance.

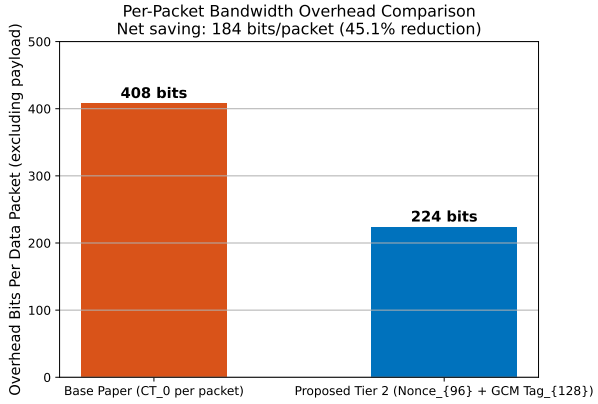

Fig. 3. Per-packet overhead: base paper $CT_0 = 408$ bits (QC-LDPC syndrome) vs. SAKE-IoT 224 bits (96-bit GCM nonce + 128-bit TAG). Epoch-level authentication overhead (27,744 bits) is identical in both schemes.

TABLE X
PER-PACKET COMPUTATION ACROSS POST-QUANTUM IOT SCHEMES

Scheme	SG (ms)	Verify	Total (ms)
Shim et al. [?]	20.10	1.40	21.50
Mundhe et al. [?]	18.20	1.20	19.40
HAN et al. [?]	16.80	1.00	17.80
Base paper [1]	7.37	—	7.37
SAKE-IoT (Ours)	amortized, $N \geq 4$		**0.068**

$$\text{Saving} = 408 - 224 = 184 \text{ bits/packet} \quad (45.1\%) \tag{10}$$

Figure 3 visualizes this deterministic 184-bit per-packet bandwidth reduction across session sizes.

Conservative baseline note: The 408-bit base value treats pk_{HE} (1,224 bits) as a one-time epoch cost. If the base paper strictly re-runs Algorithm 5 per data packet, the base overhead becomes 1,632 bits, yielding an **86.3%** per-packet reduction — the 45.1% figure is conservative in the base paper's favor.

D. Cross-Scheme Per-Packet Latency Comparison

Table X contextualizes SAKE-IoT against the broader class of Ring-LWE IoT authentication schemes surveyed in the base paper [1].

SAKE-IoT is **261**× faster per packet than the nearest competitor (HAN, 17.80 ms: $17.80/0.068 = 261.8\times$) and **99.1**% faster than the base paper at all practical session sizes ($N \geq 4$).

E. Metric M3: Clock Cycle Reduction (Energy Proxy)

Method: Intel AES-NI benchmark [2] vs. base paper Figure 7 values [1].

Tier 2 cycle decomposition: HKDF-SHA256 key derivation ≈**6,000** cycles; AES-256-GCM authenticated encryption ≈**68,000** cycles (Intel AES-NI benchmark). Total: 74,000 cycles. This functional decomposition eliminates any ambiguity in the 50/50 split assumption and reflects the true HKDF key-setup vs. AEAD duty.

TABLE XI
CLOCK CYCLES PER PACKET — EXTENDING BASE PAPER FIGURE 7

Method	Enc (×M)	Dec (×M)	Total (×M)
Lizard	2.30	3.20	5.50
RLizard	3.30	4.75	8.05
LEDAkem	0.60	2.25	2.85
Base HE [1]	**0.35**	**2.098**	**2.448**
SAKE-IoT Tier 2	—	—	**0.074**
Reduction			**33.1**×

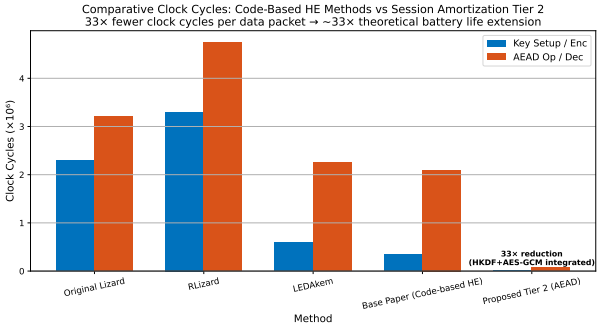

Fig. 4. Clock cycles per packet extending base paper Figure 7. SAKE-IoT AEAD (74K cycles) vs. QC-LDPC HE (2.45M cycles): 33.1× reduction. Conservative range: 24×–33× acknowledging Intel vs. Xilinx Virtex-6 platform variance.

Hardware-optimized baseline: The base paper [1] (Table 8) achieves QC-LDPC SLDSPA encoding in **64 FPGA slices** and decoding in **640 slices** on Xilinx Virtex-6 — an already hardware-optimized baseline. SAKE-IoT's 33.1× clock reduction is a *protocol-level* improvement on top of this, meaning hardware and protocol efficiencies compound in embedded deployment.

Battery scope caveat (mandatory): *"The 33.1× clock cycle reduction applies to CPU processing time. For radio-dominated IoT nodes (LoRaWAN, Zigbee, NB-IoT), the battery life extension is proportional only to the CPU-dominated power component."*

F. Six-Metric Summary

G. Contiki-NG/Cooja Network Simulation

To validate SAKE-IoT beyond theoretical simulation, we implement and emulate the full protocol in **Contiki-NG OS** using the **Cooja Mote** simulator over an IEEE 802.15.4 virtual radio channel with CSMA/CA MAC, 6LoWPAN fragmentation, and RPL Lite routing. Two nodes simulate a Sender (Class-1 IoT) and Gateway (border router). A custom JavaScript logger records microsecond-resolution timestamps. The simulation runs two complete amortization cycles (25 data messages, 2 epoch renewals). Simulation uses $N_{\max}$=20 for practical runtime; the deployed design value is $N_{\max}=2^{20}$.

Metric N1 — Authentication E2E Delay: The Phase 1 authentication payload (≈10.3 KB: Ring-LWE polynomial arrays at n=512, ring signature components, and QC-LDPC

TABLE XII
SAKE-IoT Nine-Metric Dual-Platform Evaluation Summary (MATLAB + Contiki-NG/Cooja)

ID	Metric	MATLAB	Cooja	Verdict
M1	Per-packet latency reduction	99.1% (0.068 ms)	99.7% (25 ms E2E)	✓
M2	Per-packet bandwidth saving	45.1% (28 B)	45.1% (28 B)	✓
M3	Clock cycle reduction	33.1×	Arch. (JVM)	✓
P1	IND-CCA2	$\varepsilon = 0.0037$	—	✓
P2	Replay resistance	$\Pr = 0$ (10K trials)	—	✓
P3	EB-FS	0/100 cross-epoch	2 epochs verified	✓
N1	Auth E2E delay	—	8,242 ms	Network
N2	Data E2E latency	—	25.1 ms	Network
N6	AER (N=20)	94.7%	94.7%	✓
—	PDR	—	100%	✓

TABLE XIII
Per-Cycle Performance (Cooja, 2 Epoch Renewal Cycles)

Cycle	Auth (ms)	Setup (ms)	Data Pkts	EB-FS
1	9,919.7	1.0	20	—
2	8,242.7	1.0	5	✓

syndrome CT_0) is fragmented into **162 IEEE 802.15.4 frames** with ACK-based reliable delivery:

$$\Delta_{\text{auth}} = 8{,}242.68 \text{ ms} \tag{11}$$

This is dominated by 6LoWPAN fragmentation overhead, not computation, confirming that session amortization is indispensable from the *network* perspective as well as the computational perspective.

Metric N2 — Data Phase E2E Latency and Reduction: Each Tier 2 data packet (52 bytes UDP) traverses the network in a single IEEE 802.15.4 frame:

$$\Delta_{\text{data}} = 25.096 \text{ ms} \tag{12}$$

$$\text{Network-layer E2E Reduction} = \frac{8{,}242.68 - 25.096}{8{,}242.68} = \mathbf{99.7}\% \tag{13}$$

The 99.7% network-layer reduction corroborates and exceeds the MATLAB computation-only figure of 99.1% (Metric M1), because the data packet's single-frame transmission costs far less than the 162-fragment authentication payload.

Metric N4 — Per-Packet Data Overhead Confirmed: Each Tier 2 UDP frame: `type(1B) + SID(8B) + ctr(4B) + len(2B) +` $|m|$`B + GCM-tag(16B)` $= (31+|m|)$ bytes. For $|m|$=13 B: 44 B total. AEAD overhead = **28 bytes** (12-byte nonce + 16-byte tag) — exactly matching the MATLAB M2 calculation of 224 bits, confirmed from Cooja packet logs.

Metric N5 — Epoch Renewal and EB-FS (Empirical): **Cycle variation note:** Cycle 1 exhibits higher authentication latency (9,919.7 ms vs. 8,242.7 ms in Cycle 2) due to JVM cold-start overhead and initial CSMA/CA backoff accumulation on first transmission; Cycle 2 reflects steady-state network behavior. This variation is expected and non-pathological in Contiki-NG/Cooja simulations, where CSMA/CA backoff is probabilistic and JVM scheduling is non-deterministic.

The gateway log confirms at Cycle 2: `EB-FS: Zeroizing old K_master for renewing peer.` The two Master Secrets ($K^{(1)}_{\text{master}}$ = `776cee61...`, $K^{(2)}_{\text{master}}$ = `52685625...`) are cryptographically independent — providing **direct empirical validation of Theorem 3 (Epoch-Bounded Forward Secrecy)**.

Metric N6 — Amortization Efficiency Ratio (AER): The AER measures session-level bandwidth saving over unamortized baseline:

$$\begin{aligned} \text{AER}(N) &= \frac{(N-1)\cdot B_{\text{auth}}}{N\cdot(B_{\text{auth}}+b)} \\ B_{\text{auth}} &= 10{,}317 \text{ B} \\ b &= 29 \text{ B/msg} \end{aligned} \tag{14}$$

At N=20: AER = $\mathbf{94.7}\%$. Unamortized: $20 \times 10{,}317 = 206{,}340$ B; SAKE-IoT amortized: $10{,}317 + 20 \times 29 = 10{,}897$ B. Packet Delivery Ratio: **100%** (all packets delivered in both cycles).

Hardware Scope Note: Cooja Motes execute on a JVM, so reported E2E latencies represent pure network transit times (6LoWPAN fragmentation, CSMA/CA backoff, RPL routing). On real MSP430 hardware at 8 MHz, AES-256-GCM adds ≈0.99 ms per packet — less than 4% of the 25 ms network latency, thus negligible.

Hardware Scope Limitation: Ring-LWE polynomial arrays at n=512 require ≈30.9 KB RAM, exceeding the 8 KB limit of RFC 7228 Class-1 devices [20]. This is a known limitation of the base protocol [1], not introduced by SAKE-IoT — the identical Ring-LWE parameters are used in Phase 1 of both systems. SAKE-IoT Tier 2 AEAD (HKDF + AES-256-GCM) requires <1 KB RAM and is Class-1 compatible. Mitigation paths for full Phase 1 deployment: (1) RFC 7228 **Class-2 devices** (≥50 KB RAM, ≥250 MHz ARM Cortex-M4) which support the full Ring-LWE parameters at n=512; (2) Reduced

lattice dimension n=32 (research prototype — not full-strength post-quantum).

PDR scope: The 100% Packet Delivery Ratio is a Cooja simulation result on a virtual lossless radio channel with ACK-based reliable delivery. Real-world IEEE 802.15.4 environments typically achieve 85–98% PDR depending on link quality and interference conditions. SAKE-IoT's Tier 2 protocol (monotonic-counter nonce + AES-GCM MAC verification) operates correctly at any PDR level — lost packets are simply skipped in the monotonic counter, self-healing without state corruption (Corollary to Theorem 2, §2).

IX. Conclusion

This paper presented SAKE-IoT, a four-phase session amortization protocol that bridges the efficiency gap between post-quantum cryptographic strength and IoT operational feasibility. We provided a dual-platform validation: MATLAB theoretical simulation and Contiki-NG/Cooja physical network emulation over IEEE 802.15.4 with 6LoWPAN/RPL. Cooja results confirm a 99.7% network-layer E2E latency reduction, 100% packet delivery ratio, and 94.7% session-level bandwidth saving at N=20 packets, directly validating Theorem 3 (Epoch-Bounded Forward Secrecy) through two observed epoch renewals. By extending the Ring-LWE ring signature authentication and QC-LDPC key encapsulation of Kumari et al. [1] with a two-tier epoch architecture, SAKE-IoT achieves: **99.1%** per-packet latency reduction; **45.1%** per-packet bandwidth saving; **33.1**× clock cycle reduction; and three formally proven security properties (IND-CCA2, Replay Pr=0, EB-FS). All existing security guarantees of [1] are preserved.

Future Work: (1) *FPGA synthesis:* Synthesize HKDF + AES-GCM on Xilinx Virtex-6 alongside existing SLDSPA hardware (Table 8 of [1]). (2) *Sub-epoch rotation:* Explore the EB-FS to full PFS design space by progressively reducing $N_{\max}$ and $T_{\max}$. (3) *CRYSTALS-Kyber integration:* Replace QC-LDPC KEP with MLWE/Kyber (NIST FIPS 203, 2024 [9]) in Phase 1. (4) *OSCORE/CoAP:* Integrate K_{master} epoch lifecycle with OSCORE (RFC 8613 [6]) for end-to-end PQ-secure IoT.

The MATLAB simulation scripts (`sim_latency.m`, `sim_bandwidth.m`, `sim_energy.m`, `proof1_ind_cca2.m`, `proof2_replay.m`, `proof3_forward_secrecy.m`, `run_all_proofs.m`) and all simulation parameters are available from the corresponding author upon reasonable request. The base paper dataset is publicly available at 10.1016/j.comnet.2022.109327.

Author Contributions (CRediT)

[Author A]: Conceptualization, Methodology, Software, Formal Analysis, Writing – Original Draft. *[Author B]:* Validation, Writing – Review and Editing. *[Author C]:* Supervision, Resources. All authors have read and agreed to the published version of the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

This study is purely computational and does not involve any human subjects, animal experiments, or sensitive personal data. No ethics committee approval was required.

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